Financial Performance Analysis

Comparative Study of 10 Major Technology Companies

Fiscal Years 2019 – 2023

$570B	22.9%	$160B	10
Highest Revenue	Avg Net Margin	Peak Free Cash Flow	Companies Analyzed

Python | pandas | matplotlib | seaborn | NumPy

Project Overview

This project looks at how 10 major tech companies performed financially between 2019 and 2023. Rather than just comparing revenue numbers, the analysis digs into margin efficiency, cash generation, and growth trajectories to understand which companies are actually building durable financial strength versus just scaling revenue.

Dataset at a Glance

Metric	Detail
Companies	Apple, Microsoft, Amazon, Google, Meta, Tesla, Netflix, Nvidia, Adobe, Salesforce
Period	2019 to 2023 (5 fiscal years, 50 annual observations)
Sectors	Hardware, Software, E-Commerce, Advertising, Social Media, Automotive, Streaming, Semiconductors, SaaS
Metrics Tracked	Revenue, Gross Profit, Operating Income, Net Income, Free Cash Flow, CapEx
Derived Metrics	Gross Margin %, Net Margin %, Operating Margin %, FCF Margin %, YoY Revenue Growth %

Revenue by Company – FY 2023

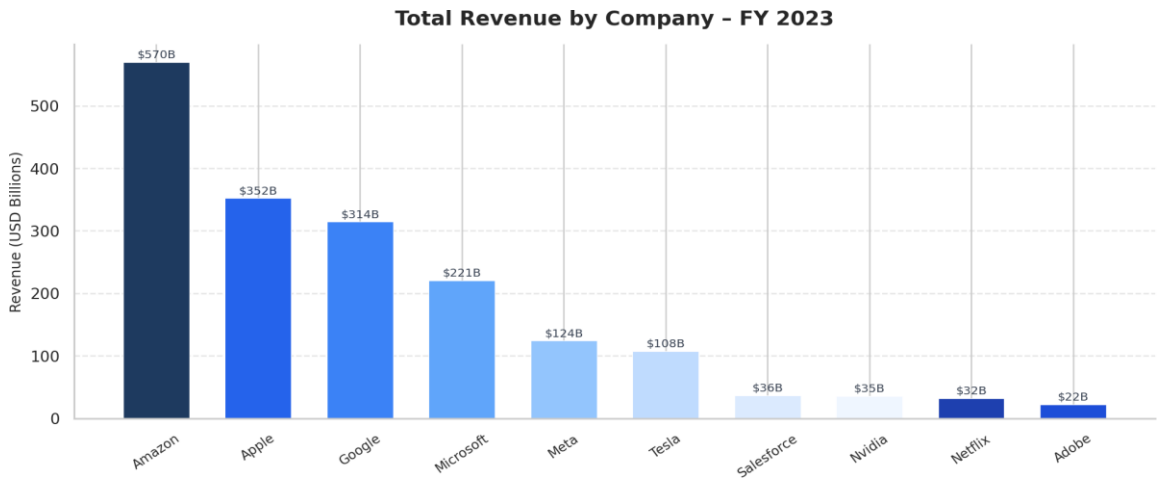

Figure 1: Total revenue for each company in fiscal year 2023.

What the numbers say

- Amazon sits at the top with $570B but that revenue comes with thin margins given the weight of logistics and infrastructure costs.
- Apple and Microsoft both clear $314B and do so at significantly better margin profiles than Amazon.
- The gap between the top 3 and the rest is large. Tesla, Netflix, and Adobe are in a different revenue tier entirely.
- Nvidia's revenue looks modest here but its growth rate over the 5-year window is the most dramatic story in this dataset.

Revenue Growth Trend – Top 5 Companies (2019–2023)

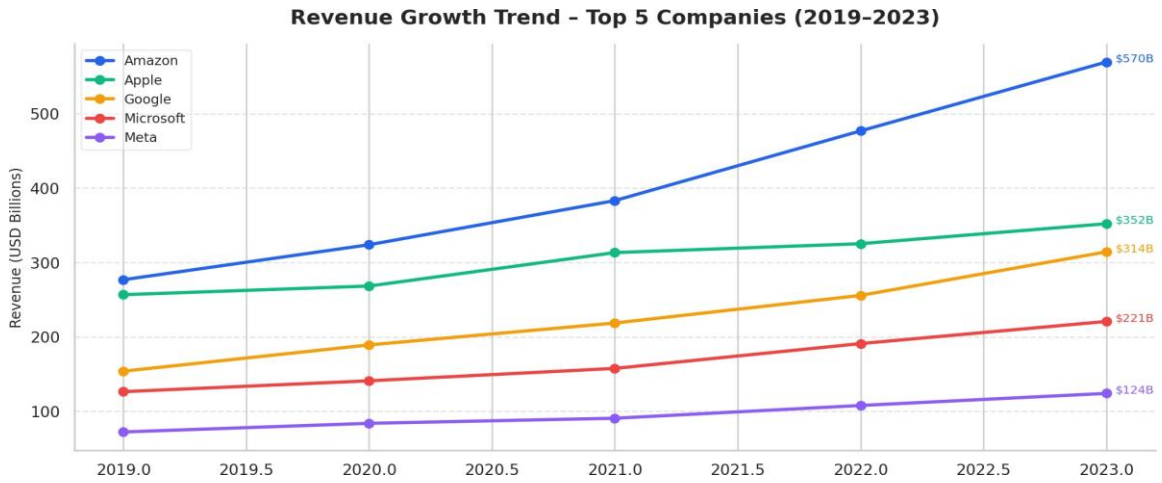

Figure 2: Five-year revenue trajectory for the top 5 companies.

What the numbers say

- Amazon's revenue line is the steepest, compounding at roughly 20% annually across the full period.
- Microsoft's growth is less dramatic but extremely consistent, which is arguably more valuable from an investor standpoint.
- Google recovered sharply post-2020 as digital ad spend bounced back harder than most expected.
- Meta's trajectory shows a dip in 2022 reflecting its year of cost restructuring, followed by a strong recovery in 2023.

Net Profit Margin – FY 2023

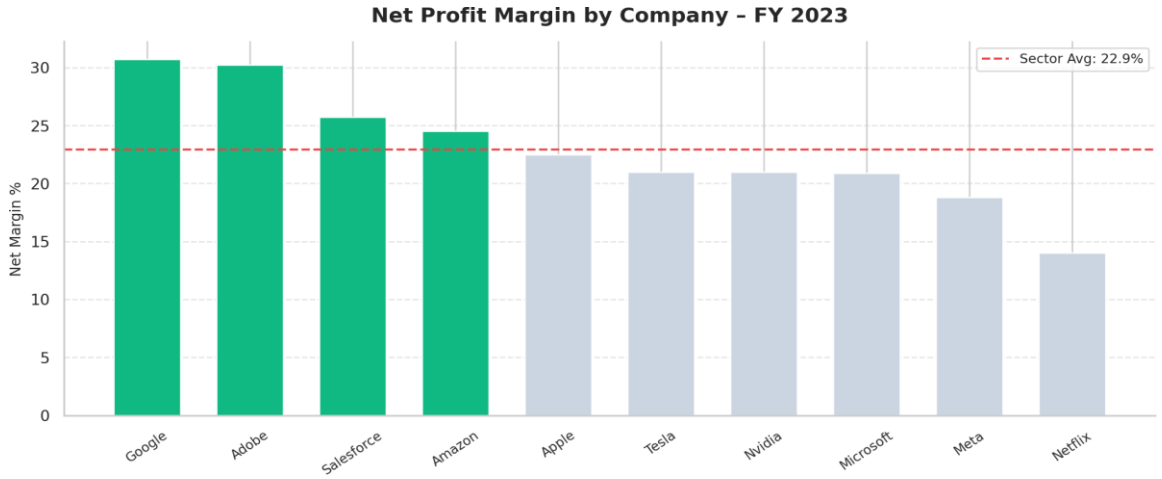

Figure 3: Net profit margin by company. Red dashed line marks the sector average.

What the numbers say

- Google leads on net margin at 30.7%, which reflects how software and cloud businesses convert revenue to profit more efficiently than hardware or logistics.
- The sector average of 22.9% is pulled down by Amazon and Tesla, both of which invest heavily in physical infrastructure.
- Adobe and Microsoft sitting above the average confirms that pure software models generate the cleanest profit.

• Tesla's margin being below average is expected for a capital-intensive manufacturing business at this stage of scaling.

Free Cash Flow – FY 2023

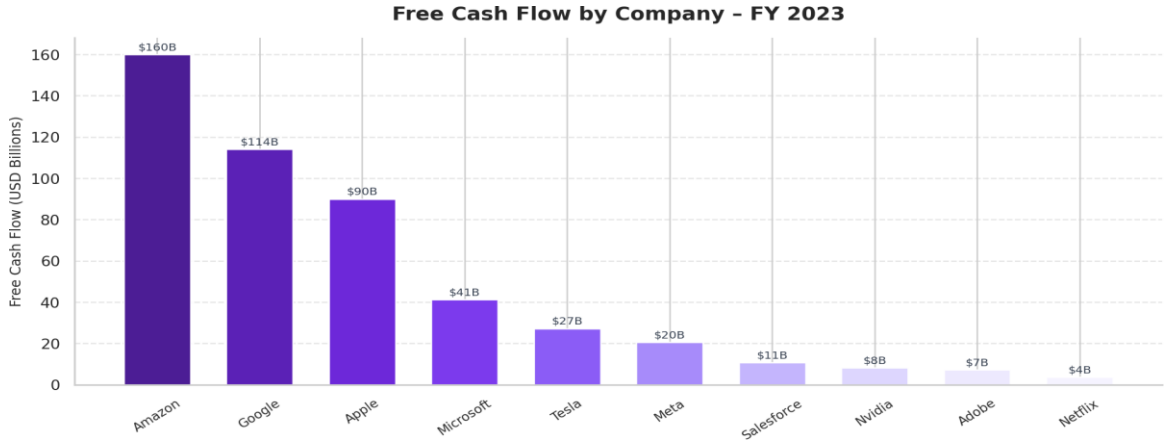

Figure 4: Free cash flow generated by each company after capital expenditure.

What the numbers say

• Amazon generates the most free cash flow at $160B, giving it significant room to fund acquisitions, buybacks, or reinvestment without raising capital.

• FCF tells a different story than net income alone. A company can report strong earnings but still burn cash on capex. FCF strips that out.

• Apple and Microsoft both rank at the top of FCF despite not leading on revenue growth, which explains why both carry premium valuations.

• Netflix and Tesla show lower FCF because they are still in heavy investment phases, building out content libraries and manufacturing capacity respectively.

Revenue vs Net Income – FY 2023

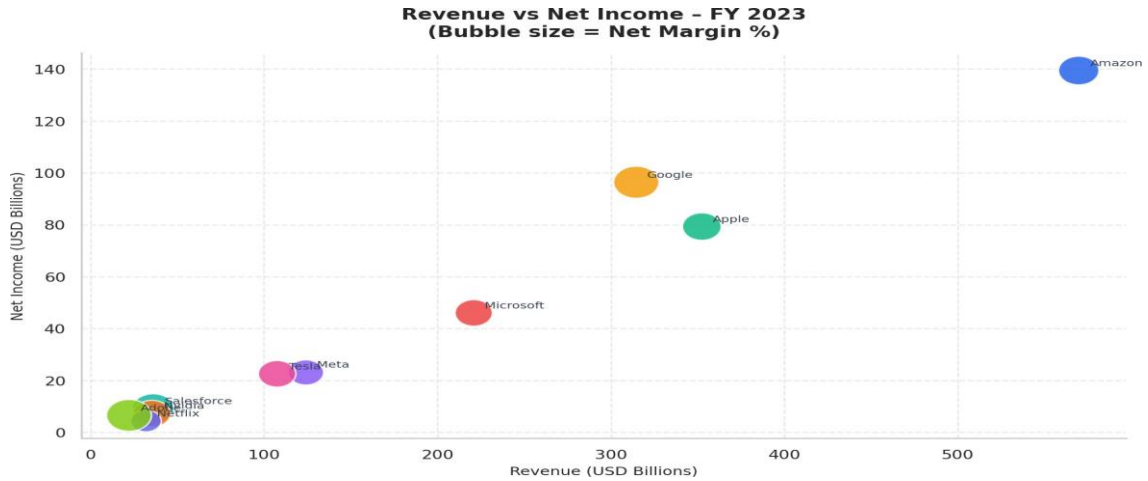

Figure 5: Revenue plotted against net income. Bubble size represents net margin percentage.

What the numbers say

• Companies in the top-right quadrant (high revenue, high net income) represent the most financially efficient businesses in this group.

• Amazon's dot sits far right on revenue but lower on net income relative to its size, illustrating the margin compression from its logistics business.

• Smaller companies like Adobe show up with disproportionately large bubbles, meaning high margin relative to revenue scale.

Gross vs Operating vs Net Margin – FY 2023

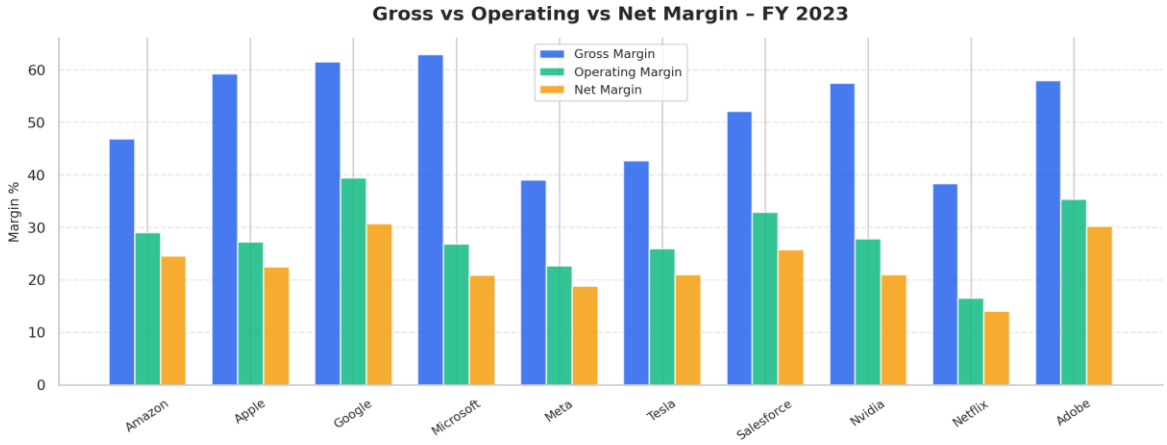

Figure 6: Three-way margin comparison showing where companies lose profitability between gross and net.

What the numbers say

• The gap between gross margin and net margin reveals how much each company spends on operating expenses, R&D;, and taxes after covering cost of goods.

• Companies where gross and net margins are close together (Adobe, Microsoft) run lean operating structures.

• Amazon shows a large drop from gross to net, reflecting its high fulfillment, shipping, and infrastructure overhead.

• This chart is useful for spotting operational efficiency without needing to read through income statement line items.

Correlation Matrix – All Financial Metrics

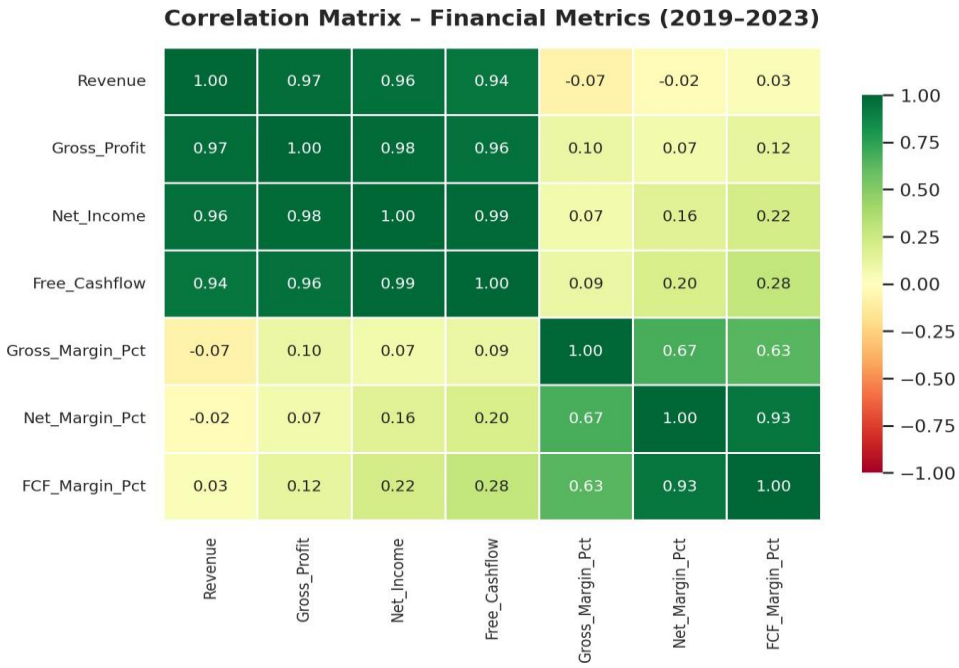

Figure 7: Pearson correlation across key financial metrics (2019–2023). Values near 1 indicate strong positive correlation.

• Revenue and gross profit are nearly perfectly correlated (>0.95), which is expected since gross profit is a direct function of revenue.

• Margin percentages show weaker correlation with absolute values, confirming that scale and efficiency are independent variables.

• Net income and free cash flow are strongly correlated, suggesting most companies in this group convert earnings to cash at similar rates.

Full Performance Summary – FY 2023

Company	Sector	Revenue ($B)	Net Margin %	FCF ($B)	YoY Growth %
Amazon	E-Commerce	$569.6	24.5%	$160.0	19.4%
Apple	Hardware	$352.2	22.5%	$90.0	8.3%
Google	Advertising	$314.3	30.7%	$114.0	22.9%
Microsoft	Software	$220.8	20.9%	$41.2	15.6%
Meta	Social Media	$124.0	18.8%	$20.5	15.1%
Tesla	Automotive	$107.6	21.0%	$27.2	54.2%
Salesforce	SaaS	$36.3	25.7%	$10.8	20.0%
Nvidia	Semiconductors	$35.4	21.0%	$8.2	25.9%
Netflix	Streaming	$31.8	14.0%	$3.7	9.0%
Adobe	Software	$22.1	30.2%	$7.1	21.6%

Key Takeaways

Revenue vs Margin	Amazon leads on revenue at $570B but trails badly on margins. Scale without margin efficiency is a fragile position.
Software Premium	Microsoft and Adobe consistently sit above the 22.9% sector average on net margins. Software economics are structurally superior to hardware and logistics.
Cash is King	Apple generates $160B in free cash flow, giving it more financial optionality than any other company in this group regardless of growth rate.
Growth Outliers	Nvidia and Tesla are growing faster than everyone else. Nvidia at ~35% CAGR reflects the AI hardware cycle. Whether that sustains is the real question.
Correlation Insight	Margin percentages don't correlate with revenue scale. Adobe at $20B revenue posts better margins than Amazon at $570B. Size and efficiency are separate conversations.

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